Mitochondrial DNA Depletion Syndrome 6 (Hepatocerebral Type) — Comprehensive Disease Dossier

Disease: Mitochondrial DNA Depletion Syndrome 6, Hepatocerebral Type (MTDPS6) **MONDO:** MONDO:0009747 · **OMIM phenotype:** 256810 · **Causal gene: MPV17** (HGNC:7224, chromosome 2p23.3) **Category:** Mendelian, autosomal recessive

Summary

Mitochondrial DNA Depletion Syndrome 6, hepatocerebral type (MTDPS6; MONDO:0009747; OMIM 256810) is a rare autosomal-recessive mitochondrial disorder caused by **biallelic loss-of-function variants in MPV17**, a gene that encodes a small hydrophobic protein of the **inner mitochondrial membrane**. The disease was originally thought to involve a peroxisomal protein, but was definitively re-localized to mitochondria and linked to hepatocerebral mtDNA depletion in 2006 (PMID: 16582910, PMID: 16582907). **Navajo neurohepatopathy (NNH)**, long described as a distinct entity among Diné (Navajo) children of the southwestern United States, is the same disease, caused by a homozygous **p.R50Q founder mutation** in MPV17 (PMID: 16909392).

Mechanistically, MPV17 is required for **mitochondrial deoxynucleotide (dNTP) homeostasis**. Loss of MPV17 lowers the mitochondrial pools of dGTP and dTTP, slowing mtDNA replication and producing **tissue-specific, liver-predominant mtDNA depletion** and consequent **oxidative phosphorylation (OXPHOS) failure** (PMID: 26760297). The protein is thought to form a non-selective channel/pore in the inner membrane that imports nucleotides (PMID: 25205723), functioning as part of a high-molecular-weight complex (PMID: 30273399). Critically, the mtDNA depletion and its downstream consequences can be **prevented or rescued by deoxynucleoside supplementation** in cellular models, nominating a rational therapy.

Clinically, most patients present in infancy with an **encephalohepatopathic** phenotype: hepatic failure/cholestasis, failure to thrive, lactic acidemia, hypoketotic hypoglycemia, and central/peripheral neurological disease, with mtDNA depletion detectable mainly in liver (PMID: 29282788). A rarer **late-onset neuromyopathic** form presents as axonal sensorimotor polyneuropathy (overlapping Charcot-Marie-Tooth) with little or no liver involvement (PMID: 30298599). Prognosis in the infantile form is poor, but **hypomorphic missense genotypes** (homozygous p.R50Q, p.P98L, p.R41Q) carry a relatively better course, and **liver transplantation** is viable in selected patients with limited neurological burden (PMID: 29282788, PMID: 32703289, PMID: 35466509). Preclinical **AAV-MPV17 gene therapy** restores liver mtDNA and OXPHOS in mouse models (PMID: 24247928).

This report is derived from **aggregated disease-level resources** (OMIM, Orphanet, MONDO) and the **primary literature** — case series, cohort studies, and mechanistic/model-organism work — rather than individual-patient EHR data.

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1. Disease Information

Overview. MTDPS6 is a Mendelian mitochondrial DNA maintenance disorder in the **mtDNA depletion syndrome (MDDS/MDS)** family. In MDDS, nuclear-gene mutations cause a severe reduction in the *quantity* of mtDNA (copy number) in affected tissues — distinct from mtDNA *point mutations* or large rearrangements. Reduced mtDNA leads to deficiency of mtDNA-encoded OXPHOS subunits and impaired energy production in the organs involved. MTDPS6 is the **hepatocerebral** form, predominantly affecting liver and the nervous system.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0009747
OMIM (phenotype)	256810
Gene	MPV17 (HGNC:7224; chr2p23.3)
ICD-10	E88.49 (other disorders of mitochondrial metabolism) / mitochondrial cytopathies
MeSH	Mitochondrial Diseases; DNA, Mitochondrial

Synonyms / alternative names. MTDPS6; MPV17-related mitochondrial DNA maintenance defect; MPV17-related hepatocerebral mitochondrial DNA depletion syndrome; MPV17 deficiency; **Navajo neurohepatopathy (NNH); Navajo neuropathy**.

Source type. Disease-level aggregated resources plus primary case series and mechanistic studies. One large review consolidates **100 affected individuals** carrying 48 MPV17 pathogenic variants (PMID: 29282788).

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2. Etiology

Primary cause — genetic. MTDPS6 is caused by **biallelic (homozygous or compound-heterozygous) pathogenic variants in MPV17**. There is no infectious or primary environmental cause. Spinazzola et al. mapped hepatocerebral MDDS to chromosome 2p21-23 and identified disease-segregating MPV17 mutations in three families; Calvo et al. independently nominated MPV17 via integrative genomics (PMID: 16582910, PMID: 16582907).

Genetic risk factors. The causal determinant is the MPV17 genotype itself. **Founder effects** elevate risk in specific populations — most notably the **p.R50Q (c.149G>A)** allele among the Navajo (Diné), and a recurrent **c.451dupC** allele reported in Japanese patients. **Consanguinity** increases risk of homozygosity, as in other autosomal-recessive disorders.

Environmental modifiers (disease triggers, not causes). Metabolic stressors that increase energetic demand or alter fuel utilization can precipitate decompensation: **fasting, intercurrent infection, and ketogenic/high-fat states**. In the Mpv17−/− mouse, a **ketogenic diet precipitates liver cirrhosis and failure** in otherwise near-asymptomatic animals (PMID: 24247928); one infant's liver function fluctuated, worsening with infection and improving with aggressive treatment (PMID: 36753038).

Protective factors. No validated protective alleles in humans. Mechanistically, **deoxynucleoside availability** is protective at the cellular level (rescues mtDNA depletion; PMID: 26760297), and avoidance of fasting/ketosis is protective at the organismal level (PMID: 24247928).

Gene-environment interaction. The clearest GxE signal is **genotype × dietary/metabolic state**: hypomorphic genotypes combined with avoidance of catabolic stress yield milder courses, whereas ketosis/fasting unmasks severe liver disease even on a permissive genetic background (mouse model, PMID: 24247928).

3. Phenotypes

The dominant presentation is **early-onset encephalohepatopathy**: "The vast majority of affected individuals presented with an early-onset encephalohepatopathic disease characterized by hepatic and neurological manifestations, failure to thrive, lactic acidemia, and mtDNA depletion detected mainly in liver tissue" (PMID: 29282788).

Phenotype	Type	Onset	Frequency / notes	Suggested HPO
Hepatic failure / liver dysfunction	Clinical sign	Neonatal–infantile	Core feature; ALF or progressive cholestasis	HP:0001399 Hepatic failure
Cholestasis / jaundice	Lab + sign	Neonatal–infantile	91.3% (21/23) in Japanese cohort	HP:0001396 Cholestasis
Failure to thrive	Sign	Infantile	56.5% (13/23); most common initial manifestation	HP:0001508 Failure to thrive
Lactic acidemia / lactic acidosis	Lab abnormality	Neonatal–infantile	Core biochemical feature	HP:0003128 Lactic acidosis
Hypoketotic hypoglycemia	Lab abnormality	Infantile	Common; triggered by fasting/infection	HP:0012418 Hypoglycemia
Peripheral neuropathy (sensorimotor axonal)	Sign	Childhood–adult (late form)	Hallmark of neuromyopathic form / NNH	HP:0003477 Peripheral axonal neuropathy
Leukoencephalopathy / CNS involvement	Imaging/sign	Variable	Cerebral white-matter disease in NNH	HP:0002352 Leukoencephalopathy
Corneal anesthesia / scarring	Sign	NNH forms	Characteristic of NNH	HP:0000532 Corneal opacity
Acral mutilation / self-injury	Sign	NNH (sensory loss)	NNH feature from sensory neuropathy	HP:0009775 Acral ulceration
Hepatocellular carcinoma	Complication	Childhood	Reported complication	HP:0001402 Hepatocellular carcinoma
Myopathy	Sign			

Phenotype	Type	Onset	Frequency / notes	Suggested HPO
		Late-onset form	Neuromyopathic form	HP:0003198 Myopathy
Developmental delay / hypotonia	Sign	Infantile	In hepatocerebral form	HP:0001263 / HP:0001252

Severity, progression, QoL. The infantile form is **severe and progressive**, frequently fatal before age 1–2 years; recurrent metabolic acidosis with intercurrent infection is characteristic of NNH (PMID: 16909392). Quality of life is profoundly impaired by liver failure, feeding difficulties, developmental delay, and (in NNH/late form) disabling neuropathy and acral mutilation. The late-onset neuromyopathic form is milder with slower progression and preserved liver function (PMID: 30298599).

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4. Genetic / Molecular Information

Causal gene. MPV17 (HGNC:7224; chr2p23.3), encoding a hydrophobic **inner mitochondrial membrane** protein. "MPV17 is a mitochondrial inner membrane protein, and its absence or malfunction causes oxidative phosphorylation (OXPHOS) failure and mtDNA depletion" (PMID: 16582910).

Pathogenic variants. The El-Hattab review catalogues **48 MPV17 pathogenic variants across 100 individuals, ~half missense** (PMID: 29282788). Representative variants:

Variant (cDNA / protein)	Type	Significance	Population / note
c.149G>A / p.R50Q	Missense	Pathogenic	Navajo founder; milder prognosis
c.293C>T / p.P98L	Missense	Pathogenic	Associated with better prognosis
c.122G>A / p.R41Q	Missense	Pathogenic	Axonal neuropathy / late-onset form
c.451dupC	Frameshift	Pathogenic	Recurrent in Japanese patients
c.55delC / p.Gln19ArgfsTer3	Frameshift	Pathogenic (novel)	Infantile liver failure
c.277C>T / p.Gln93*	Nonsense	Pathogenic	Infantile hepatic failure
c.461+1G>C	Splice-site	Pathogenic (novel)	Hepatic failure
c.376-9T>G	Near-splice	Pathogenic	In-frame 11-aa deletion; neuropathy
c.388G>C / p.Ala130Pro	Missense	Likely pathogenic	Infantile series

Variant classification. Classified per ACMG/AMP; most are pathogenic/likely pathogenic. **Allele frequency** in gnomAD is very low (rare), consistent with a recessive, severe disorder (specific frequencies not independently extracted here).

Functional consequence. Loss of function (reduced/absent dNTP transport), whether by truncation, splice disruption, or destabilizing/complex-disrupting missense changes. Missense alleles still localize correctly to mitochondria but fail to join a high-MW complex (PMID: 30273399).

Origin. Germline, biallelic. No somatic/oncogenic role (the HCC complication is secondary to chronic liver injury, not a somatic driver).

Modifier genes / epigenetics / chromosomal abnormalities. No established modifier genes, disease-specific epigenetic signatures, or large chromosomal abnormalities. Severity is driven primarily by the **specific MPV17 genotype** (hypomorphic missense vs. null).

5. Environmental Information

- **Environmental/toxic factors:** none causative. Metabolic stressors (fasting, catabolism) act as **decompensation triggers**.
- **Lifestyle/dietary factors: ketogenic/high-fat dietary states** are deleterious in the mouse model (PMID: 24247928); fasting and infection worsen infantile disease (PMID: 36753038).
- **Infectious agents:** none cause the disease, but **intercurrent infections** precipitate metabolic crises and liver decompensation (PMID: 16909392).

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6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic MPV17 loss-of-function variants → absent or non-functional MPV17 protein** in the inner mitochondrial membrane (missense alleles localize but fail to assemble into the high-MW complex). *(demonstrated — PMID: 16582910, PMID: 30273399)*
2. **Loss of the MPV17 channel/pore → impaired import/homeostasis of deoxynucleotides** across the inner membrane. *(inferred channel function — PMID: 25205723)*
3. → **Mitochondrial dNTP insufficiency** (decreased dGTP and dTTP in liver mitochondria). *(demonstrated — PMID: 26760297)*
4. → **Slowed mtDNA replication** by the mtDNA replisome (POLG) due to inadequate substrate supply. *(demonstrated — PMID: 26760297)*
5. → **Tissue-specific mtDNA depletion**, most severe in **liver** (where salvage-pathway dNTP supply cannot compensate), relatively spared in tissues with adequate dNTP pools (kidney, brain in mouse). *(demonstrated — PMID: 26760297)*
6. → **Deficiency of mtDNA-encoded OXPHOS subunits → respiratory-chain/OXPHOS failure** and reduced ATP. *(demonstrated — PMID: 16582910)*

Branch A (hepatic): OXPHOS failure in hepatocytes → impaired energy metabolism, cholestasis, steatosis, necrosis, fibrosis/cirrhosis → **liver failure**; chronic injury → **hepatocellular carcinoma** risk. *(demonstrated — PMID: 37204315, PMID: 22137549)*

Branch B (systemic metabolic): OXPHOS failure → **lactic acidemia**, **hypoketotic hypoglycemia**, **failure to thrive**; impaired mitochondrial AdoMet utilization → **hypermethioninemia and elevated S-adenosylmethionine**. *(demonstrated — PMID: 29282788, PMID: 22137549)*

Branch C (neurological): energy failure in CNS and peripheral nerve → **leukoencephalopathy, developmental delay, axonal sensorimotor polyneuropathy, corneal anesthesia, acral mutilation**. *(demonstrated clinically; neuronal model — PMID: 36587049)*

1. **Metabolic trigger (ketosis/fasting/infection)** modulates branch severity → precipitates acute decompensation and liver crisis. *(demonstrated in mouse — PMID: 24247928)*

Detail by category

- **Biochemical core defect:** mitochondrial **dNTP insufficiency** → "our study establishes dNTP insufficiency in the mitochondria as the cause of mitochondrial DNA depletion in MPV17 deficiency, and identifies deoxynucleoside supplementation as a potential therapeutic strategy" (PMID: 26760297).
- **Protein dysfunction:** MPV17 forms a **channel/pore** ("the Mpv17 protein forms a channel in the inner mitochondrial membrane, allowing small molecules - in vertebrates probably nucleotides"; PMID: 25205723) and participates in a **high-molecular-weight complex**; pathological missense alleles compromise complex formation (PMID: 30273399).
- **Metabolic changes:** OXPHOS failure, lactic acidosis, hypoglycemia, elevated methionine/AdoMet.
- **Cellular processes:** impaired mtDNA replication, bioenergetic failure, oxidative stress and elevated ROS (shown in Drosophila dMpv17 knockdown; PMID: 36587049), hepatocyte injury/necrosis and fibrosis.
- **Tissue damage mechanisms:** energy-deprivation-driven hepatocyte death, cholestasis, fibrosis/cirrhosis; axonal degeneration in peripheral nerve.
- **Immune involvement:** none primary.

Suggested GO / CL / anatomy terms. GO:0006264 mitochondrial DNA replication; GO:0006119 oxidative phosphorylation; GO:1990542 mitochondrial transmembrane transport; GO:0005743 mitochondrial inner membrane (CC); GO:0005739 mitochondrion (CC). Cell types: **hepatocyte (CL:0000182)**, neuron/peripheral sensory & motor neuron (CL:0000540).

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7. Anatomical Structures Affected

- **Primary organ: liver (UBERON:0002107)** — hepatocytes bear the greatest mtDNA depletion and OXPHOS failure.
- **Nervous system: brain/CNS white matter (UBERON:0002316)** and **peripheral nerves (UBERON:0001021)**; cornea (UBERON:0000964) in NNH.
- **Secondary / systemic:** growth failure (failure to thrive), systemic lactic acidosis; kidney involvement reported in the mouse model (glomerulosclerosis) and occasionally clinically.
- **Body systems:** digestive/hepatobiliary, central and peripheral nervous, metabolic/endocrine (glucose homeostasis).
- **Tissue types:** hepatic epithelial parenchyma; nervous tissue (CNS myelin, peripheral axons); muscle (neuromyopathic form).
- **Subcellular compartment: mitochondrion / mitochondrial inner membrane** (GO:0005743) — the primary site of the lesion.
- **Lateralization:** systemic/bilateral; neuropathy is typically length-dependent and symmetric.

8. Temporal Development

- **Onset:** Most commonly **neonatal to early-infantile** (encephalohepatopathic form). In the single-centre series, MPV17 liver involvement developed at a **median ~2.5 months** (PMID: 37204315). The **late-onset neuromyopathic** form presents in **juvenile-to-adult** years (PMID: 30298599).
- **Pattern:** Acute liver failure or subacute/progressive cholestasis in infants; chronic insidious progression in the late form.
- **Course:** Infantile form is **rapidly progressive**; many die within months of presentation (median death ~8 months; 5/10 MPV17 patients died at median 8 months in one series, PMID: 37204315). NNH classically has **infantile, childhood, and classic** sub-forms (PMID: 16909392).
- **Critical period / intervention window:** Early infancy — before irreversible neurological injury — is the key window; pre-transplant neurological burden strongly predicts outcome.
- **Remission:** No spontaneous remission; stabilization is treatment-dependent (dietary management, transplant).

9. Inheritance and Population

- **Inheritance: Autosomal recessive.** Biallelic MPV17 variants; carriers are asymptomatic.
- **Penetrance / expressivity:** High penetrance in biallelic carriers; **variable expressivity** strongly influenced by genotype (null vs. hypomorphic missense).
- **Founder effects: p.R50Q among the Navajo (Diné)** — "Identification of a single missense mutation in patients with NNH confirms that the disease is probably due to a founder effect" (PMID: 16909392); recurrent **c.451dupC** in Japanese patients (PMID: 32703289).
- **Consanguinity:** Increases homozygosity risk, relevant in several reported families.
- **Epidemiology:** Ultra-rare globally (Orphanet rare-disease classification; exact prevalence/incidence not established). NNH is comparatively more prevalent among Navajo children of the southwestern USA.
- **Sex ratio:** No strong sex predilection (autosomal recessive).
- **Geographic/variant distribution:** p.R50Q — southwestern USA (Navajo); c.451dupC — Japan; others distributed worldwide.

10. Diagnostics

Laboratory / biochemistry. Elevated lactate (lactic acidemia), hypoketotic hypoglycemia, abnormal liver function tests, cholestasis, and characteristically **marked hypermethioninemia with elevated S-adenosylmethionine** — "plasma methionine concentrations as high as 15-21-fold above the upper limit... S-adenosylmethionine (AdoMet) 4.4-8.6-fold above the upper limit" (PMID: 22137549). Hypermethioninemia/cholestasis frequently causes **initial misdiagnosis as tyrosinemia** (PMID: 31664948).

Tissue mtDNA quantification. Reduced mtDNA copy number in liver is the diagnostic hallmark; quantitative mtDNA analysis on liver tissue is central (PMID: 28209105). Respiratory-chain enzymology may be abnormal but is variably sensitive.

Histopathology (liver biopsy). Variable **necrosis, steatosis, cholestasis, and fibrosis/cirrhosis** (PMID: 37204315).

Genetic testing (definitive). Whole-exome/whole-genome or targeted MDDS gene-panel sequencing (including MPV17, DGUOK, POLG, TK2, FBXL4) confirms diagnosis; NGS "should be considered the first-tier diagnostic test obviating the need for more invasive testing like muscle biopsies" (PMID: 35750291). Single-gene MPV17 testing is appropriate where a founder allele is suspected (e.g., Navajo ancestry → p.R50Q). **Rapid genetic testing is advised** in infantile ALF to guide transplant decisions (PMID: 37204315).

Differential diagnosis. Tyrosinemia type I, other hepatocerebral MDDS (**DGUOK, POLG**), galactosemia, neonatal hemochromatosis, bile-acid synthesis defects, and other causes of infantile acute liver failure. Distinguishing feature: liver mtDNA depletion + MPV17 variants.

Screening. No population newborn screening exists. **Carrier and cascade screening** is valuable in founder populations (Navajo p.R50Q); prenatal/preimplantation testing is available once the familial variant is known.

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11. Outcome / Prognosis

- **Survival/mortality:** Poor in the infantile hepatocerebral form. Representative mortality: **17/24 MDDS patients died (median age 8 months)**, including **5/10 MPV17 at median 8 months** (PMID: 37204315).
- **Genotype-dependent prognosis:** "A genotype with biallelic missense variants, in particular homozygous p.R50Q, p.P98L, and p.R41Q, can carry a relatively better prognosis" (PMID: 29282788).
- **Transplant outcomes:** Of 12 transplanted Japanese patients, "five… survived; while seven who had remarkable neurological symptoms before LT died" (PMID: 32703289); survivors were enriched for p.R50Q/p.P98L. Three MPV17 patients in a UK series were alive 3–19 years post-LT (PMID: 37204315). "MPV17 deficiency should not be considered a contraindication to liver transplantation" (PMID: 35466509).
- **Complications:** Progressive liver failure, **hepatocellular carcinoma** ("At age 7 years he was found to have hepatocellular carcinoma, underwent a liver transplant and died of progressive liver and renal failure at age almost 9 years"; PMID: 22137549), progressive neurological disease (which may continue after LT), renal involvement.
- **Prognostic factors:** Genotype (missense/hypomorphic → better), **pre-transplant neurological burden** (worse → post-LT mortality), degree of liver failure.

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12. Treatment

No curative pharmacotherapy is approved; management is largely supportive.

Intervention	Rationale / evidence	NCIT suggestion
Avoid fasting/ketosis; frequent feeds, slow-release carbohydrate, **MCT-based diet**	Prevents catabolic decompensation; survivors managed with MCT diet + supplements (PMID: 36753038)	NCIT:C15227 Dietary Intervention
Deoxynucleoside / deoxynucleotide supplementation (investigational)	Rescues mtDNA depletion in patient fibroblasts (PMID: 26760297)	NCIT:C1505 Dietary Supplement
CoQ10, carnitine, vitamin cofactors	Supportive mitochondrial care; used in survivors (PMID: 36753038)	NCIT:C1505
Liver transplantation	Viable in selected patients with limited neurology (PMID: 35466509, PMID: 32703289, PMID: 37204315)	NCIT:C15393 Liver Transplantation
Management of hypoglycemia, acidosis, infection	Acute metabolic crisis control (PMID: 16909392)	NCIT:C15329 Supportive Care
AAV-MPV17 gene therapy (preclinical)	"restore liver mtDNA copy number and oxidative phosphorylation (OXPHOS) proficiency, and prevent liver failure induced by the KD" (PMID: 24247928)	NCIT:C15431 Gene Therapy
Drug repurposing (yeast-identified dNTP-pool modulators)	"we identify ten drugs active against MPV17 disorder, modelled in yeast" (PMID: 34830106)	—

Pharmacogenomics / personalized medicine: Genotype guides prognosis and transplant candidacy (hypomorphic missense favorable). **Avoid sodium valproate** and other mitochondrial toxins — valproate precipitated liver injury in POLG patients in the same series and is a general caution in MDDS (PMID: 37204315). No approved RNA/cell therapies.

13. Prevention

- **Primary prevention:** None for occurrence (genetic); **genetic counseling** and **carrier/cascade screening** in founder populations (Navajo p.R50Q) reduce incidence; prenatal and preimplantation genetic testing available for known familial variants.
- **Secondary prevention:** Early genetic diagnosis in infantile ALF to enable timely dietary management and transplant evaluation; rapid testing advised (PMID: 37204315).
- **Tertiary prevention:** Avoid fasting/ketosis and hepatotoxins (valproate); aggressively treat intercurrent infections; monitor for HCC in survivors; neurological surveillance.
- **Behavioral/dietary interventions:** Structured feeding regimens, MCT diet, avoidance of prolonged fasting.
- **Immunization / public health / environmental interventions:** Standard childhood immunization to reduce infection-triggered crises; otherwise not applicable.

14. Other Species / Natural Disease

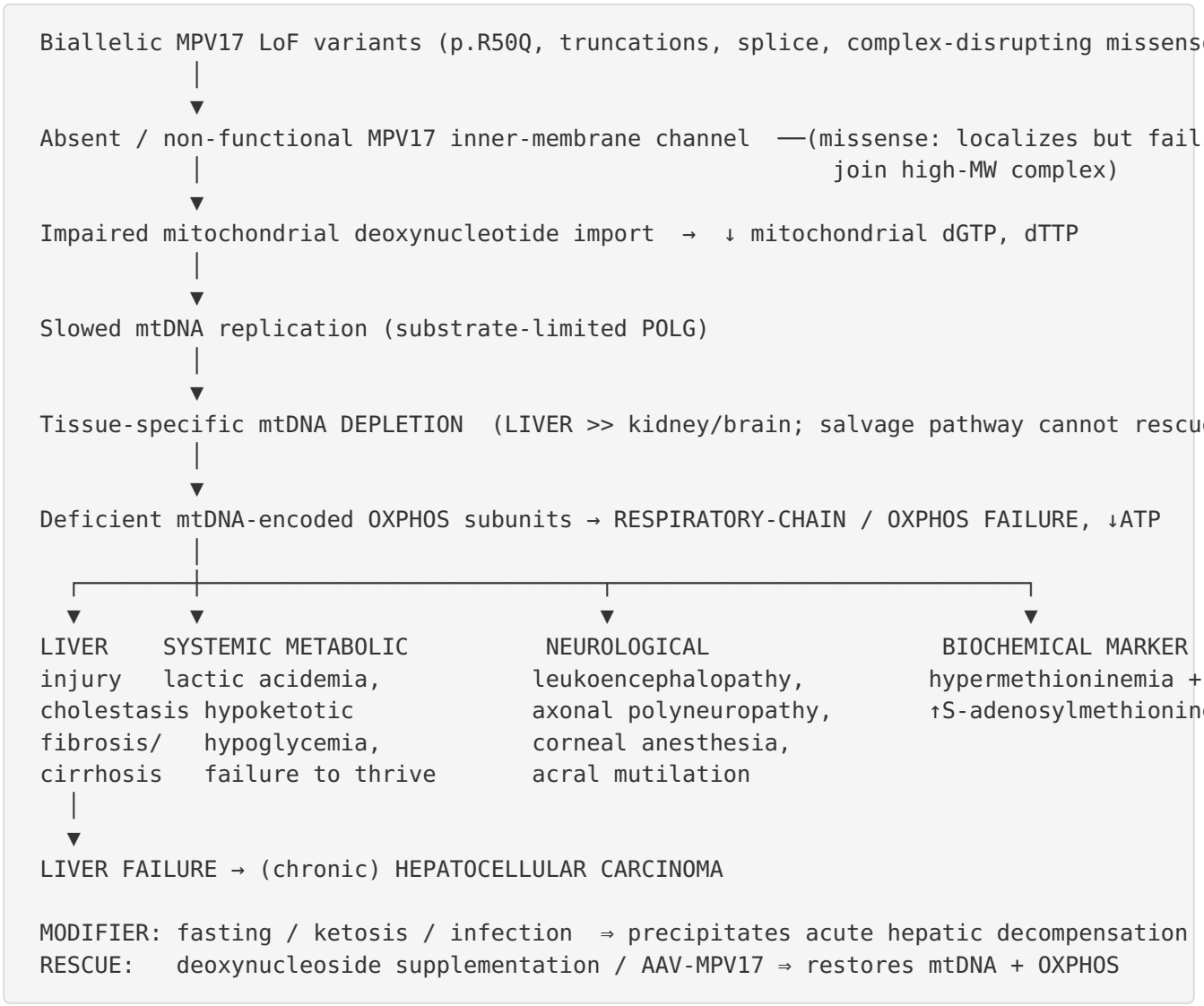
- **Taxonomy / orthologs:** Human **MPV17**; mouse **Mpv17** (NCBI Gene 17527); dog **MPV17**; *Drosophila* **CG11077 (dMpv17)**; yeast ortholog **SYM1** (*Saccharomyces cerevisiae*).
- **Natural disease in animals:** A **canine MPV17 truncation** (1-bp insertion in exon 4, enriched in Boxers) exists but is **clinically silent** — "despite the truncation of MPV17, we could not find any common phenotypes in the genetically affected dogs," suggesting species-specific compensation (PMID: 26353863).
- **Comparative biology:** Mouse Mpv17−/− shows renal (glomerulosclerosis) and hearing defects not prominent in humans, and is diet-dependent for liver disease (PMID: 24247928, PMID: 25205723). The dNTP-import function is **evolutionarily conserved** from yeast (SYM1) to vertebrates.
- **Zoonotic potential:** None (genetic disease).

15. Model Organisms

Model	Type	Key features	Limitation	Reference
Mpv17−/− mouse	Mammalian KO	Near-asymptomatic on standard diet; **ketogenic diet → cirrhosis/liver failure**; renal + hearing defects	No spontaneous severe liver phenotype; extra renal/auditory features	PMID: 24247928, PMID: 25205723
AAV-MPV17 mouse (gene therapy)	Rescue model	Liver-specific human MPV17 restores mtDNA/ OXPHOS, prevents KD-induced failure	Liver-directed only	PMID: 24247928
Patient fibroblasts (quiescent)	In vitro	Absent protein, decreased dNTPs, mtDNA depletion; **rescued by deoxynucleosides**	Cell-autonomous, non-hepatic	PMID: 26760297
***Drosophila* dMpv17 knockdown**	Invertebrate	Recapitulates **neurological defects**: impaired locomotion/ learning, ↓mtDNA/ATP, ↑lactate/ROS, abnormal NMJ	Not hepatic	PMID: 36587049
Yeast SYM1 (sym1 mutants)	In vitro / genetic	OXPHOS + mtDNA instability; models missense alleles; **drug-repurposing platform** (10 active molecules)	Unicellular; no tissue phenotype	PMID: 30273399, PMID: 34830106
Canine MPV17 truncation	Natural large-animal	Clinically silent; model of species-specific compensation	No disease phenotype	PMID: 26353863

Applications: The mouse enables gene-therapy and diet-modulation studies; fibroblasts validate deoxynucleoside rescue; *Drosophila* uniquely captures the neurological phenotype absent in mouse/zebrafish/human-cell models; yeast enables mechanistic dissection and drug screening.

Mechanistic Model (synthesis)



Evidence Base

PMID	Role in this dossier
16582910	Identifies MPV17 as causal gene; inner-membrane localization; OXPHOS/mtDNA depletion
16582907	Independent integrative-genomics discovery of MPV17
26760297	Core mechanism: dNTP insufficiency; deoxynucleoside rescue
25205723	MPV17 channel function; cross-species model comparison
30273399	Missense alleles disrupt high-MW complex (yeast SYM1)
29282788	100-patient review; phenotype spectrum; genotype-prognosis
16909392	Navajo neurohepatopathy = MPV17 R50Q founder disease
32703289	Japanese cohort; transplant outcomes; phenotype frequencies
35466509	LT not a contraindication in MPV17/NNH
37204315	UK MDDS series; mortality; MPV17 LT survivors
24247928	Mouse KD-dependence; AAV gene-therapy proof of concept
34830106	Yeast drug-repurposing; dNTP-pool modulators
36587049	Drosophila model of neurological phenotype
30298599	Late-onset axonal polyneuropathy form (p.R41Q)
22137549	HCC complication; hypermethioninemia/↑AdoMet
31664948	Tyrosinemia misdiagnosis; novel splice/nonsense variants
36753038	Infantile series; MCT-diet management; infection triggers
35750291	WES as first-tier diagnostic; novel variants
28209105	NNH clinicopathologic diagnosis; liver mtDNA analysis
26353863	Clinically silent canine MPV17 truncation

Limitations and Knowledge Gaps

1. **Precise epidemiology unknown.** No reliable prevalence/incidence figures; counts derive from case series and a 100-patient review, subject to ascertainment bias.
2. **Exact molecular function incompletely resolved.** MPV17 is proposed as a dNTP-importing channel within a high-MW complex, but the complex composition, selectivity, and regulation remain uncharacterized.
3. **Tissue-specificity of liver predominance** is attributed to salvage-pathway differences but is not fully mechanistically explained.
4. **Human therapeutic evidence is thin.** Deoxynucleoside supplementation and AAV gene therapy are supported only by in vitro/mouse data; no human trials with NCT identifiers were identified.
5. **Genotype-phenotype correlations are coarse** — missense/hypomorphic alleles trend milder, but individual variability (expressivity) is incompletely predicted.
6. **Allele frequencies** in gnomAD were not independently extracted here for each variant.
7. **Neurological progression after liver transplant** is recognized but poorly quantified longitudinally.

Proposed Follow-up Experiments / Actions

1. **Human deoxynucleoside/deoxynucleotide trial** in MPV17 patients (building on fibroblast rescue, PMID: 26760297) with mtDNA copy number and liver-function endpoints.
2. **AAV-MPV17 translational development** toward IND, including biodistribution, durability, and whether liver-directed therapy mitigates neurological disease (PMID: 24247928).
3. **Structural characterization** of the MPV17 high-MW complex (cryo-EM/AlphaFold-Multimer) to define the channel and the effect of missense alleles (PMID: 30273399).
4. **Prospective natural-history registry** capturing genotype, onset, transplant timing, and neurological trajectory to refine prognostic modeling and the critical intervention window.
5. **Validate yeast-nominated drugs** (PMID: 34830106) in patient fibroblasts and the Drosophila neurological model (PMID: 36587049).
6. **Carrier-screening program evaluation** in the Navajo population for the p.R50Q founder allele, with genetic-counseling outcomes.
7. **Biomarker study** of plasma methionine/AdoMet (PMID: 22137549) as a non-invasive diagnostic/monitoring tool and to reduce tyrosinemia misdiagnosis (PMID: 31664948).

Evidence source types: human clinical (case series, cohorts, reviews), model organism (mouse, Drosophila, yeast, dog), in vitro (patient fibroblasts), and computational/comparative genomics. All mechanistic and clinical claims are cited to primary literature by PMID.